

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

*I K. Akshita (192312522) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: K. Akshita

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NS

Date
25/11/20

CSA1714 - Artificial Intelligence

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CO1 - AT3 - Descriptive Test

① SMART HOME AUTOMATION SYSTEM

Ans: Introduction:- A Smart Home Automation system is an AI-based system that automatically controls home appliances such as lighting, temp, and security devices. It uses sensors like motion detectors, temp sensors, and smart locks to monitor the environment & make intelligent decisions.

Types of Intelligent Agents

1. Simple Reflex Agent

A Simple Reflex Agent makes decisions only based on the current sensor input using predefined IF-THEN rules.

Working in Smart Home

- IF motion is detected → turn ON the light
- IF no motion is detected → turn OFF the light.

Advantages

- Fast response
- Simple design

Limitations

- Does not remember previous events.
- Cannot learn user preferences.

2. Model-Based Reflex Agent

A Model-Based Agent maintains an internal model of the environment & stores previous information to make decisions.

Working in Smart Home

- Remembers room occupancy.

→ Tracks previous temp values.

Advantages

→ Uses memory

→ Performs well in partially observable environments.

Limitations

→ More complex than a simple reflex agent.

→ Requires additional memory.

3. Goal-Based Agent :-

A Goal-Based Agent chooses actions that help achieve specific objectives.

Working :-

If the room temp becomes 34°C , the agent switches on the air condition until the desired temp (24°C) is reached.

Advantages :-

→ Goal-oriented decision making

→ can evaluate different actions.

Limitation :-

→ Requires planning

→ Decision making may take more time.

Justification

→ It remembers previous events.

→ It learns user preferences.

Conclusion :-

A Hybrid Intelligent Agent provides better automation than individual agent types because it combines memory, planning and optimization.

Robot in an Unknown MazeIntroduction:-

A robot placed in an unknown maze must find the exit without prior knowledge of the environment. AI search algorithms help the robot explore the maze & identify the best path while avoiding obstacles.

* Uniform cost Search (UCS)

Definition:- Uniform cost search is an uninformed search algorithm that expands the node with the lowest cumulative path cost first.

Working

- starts from the initial position
- calculates the total path cost
- Expands the least-cost node
- Continues until the exit is found.

Advantages

- Complete algorithm
- Finds the minimum-cost path.

Disadvantages

- Requires large memory
- High execution time for large search spaces.

Complexity:-

- Time complexity: $O(b^{c^*} \epsilon)$
- Space complexity: $O(b^{c^*} \epsilon)$

b = Branching factor

c^* = cost of optimal solution

ϵ = minimum step cost.

Iterative Deepening Search (IDS):-

Iterative deepening search repeatedly performs Depth-limited Search with increasing depth until the goal is found.

Working

- Search depth = 0
- Search depth = 1
- Search depth = 2
- Continue increasing depth until the exit is reached.

Advantages

- Low memory usage
- Complete algorithm.

Disadvantages

- Repeats node expansion
- Slightly slower than BFS.

Complexity

- Time Complexity : $O(b^d)$
- Space Complexity : $O(b^d)$

b = Branching factor
 d = Depth of Solution.

Relation to Intelligent Agents

* Simple Reflex Agent

Acts only on current sensor input & cannot remember previously visited locations.

* Goal-Based Agent

Plans movements to reach the target exit.

Conclusion:-

Uniform cost search is optimal for finding the least-cost path, while Iterative Deepening search is memory efficient. Combining a Model-Based-Goal-Based Agent with UCS provides the best navigation strategy for an unknown maze.